Portugal
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Guilherme Lopes

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Work Experience

Protocol Engineer **May 2025 - February 2026**
Somnia Foundation / Virtual Society Foundation

- **Contributed** broadly to the core C++ L1 node by resolving data encoding bugs, writing core RPC API endpoints, and engineering EVM compiler optimizations that significantly increased execution speed.
- **Optimized** database internals for heavy workloads—specifically improving data compression during large state writes—to significantly increase performance and query speeds.
- **Architected** the native oracle system from scratch, building C++ off-chain worker nodes and Solidity smart contracts to fetch external GET/POST requests and LLM data on-chain.

Software Engineer **June 2024 - May 2025**
Community Labs / Astro Labs

- **Architected** and built a cross-chain stablecoin bridge connecting Ethereum, Arweave, and AO networks to securely process multi-network asset transfers.
- **Implemented** core bridge components using Solidity smart contracts and oracle infrastructure to facilitate decentralized asset bridging.
- **Secured** cross-chain message validation by deploying MPC-based signing nodes (C++) on Trusted Execution Environment (TEE) hardware to prevent replay attacks and ensure key integrity.

Software Engineering Intern **June 2023 - September 2023**
Nethermind

- **Contributed** to the Starknet Go client library by developing and fixing automated tests, ensuring system reliability and code quality.

Blockchain Researcher **September 2022 - September 2023**

- **Researched** and implemented multiple consensus protocols (e.g., Proof of Work, Proof of Authority) for a master's thesis (research grant awarded).
- **Developed** a modular consensus library and a live-testing tool in OCaml to easily integrate and evaluate the performance (TPS, block times) of custom protocols on a Tezos node.

Software Engineer (Freelance / Contract) **October 2019 - September 2022**

- **Managed** an independent freelancing business, collaborating directly with clients to define technical requirements and deliver end-to-end software solutions.
- **Developed** and deployed full-stack web applications, Discord bots, and internal automation tools (e.g., ticket systems, inventory management) using Rust, Go, and Python.
- **Engineered** Solana smart contracts for early-stage startups using Rust and the Anchor framework.

Education

Master of Science (MSc) in Computer Science **2021 - 2023**
Blockchain, Consensus Protocols, Compilers
University of Beira Interior (Top 3 in class)

Bachelor of Science in Computer Science **2018 - 2021**
Distributed Systems (Best in Class)
University of Beira Interior

Relevant Projects

Arrow Distributed Directory Protocol and Distributed Database / Filesystem
- Implemented a distributed directory protocol node in Go with a live visualization tool for network state and access control, and later created a distributed database and filesystem using Rust and Libp2p to enhance efficient data storage and retrieval in a peer-to-peer network.

Ethereum (Solidity) & Solana (Rust / Anchor)
- Built EVM and Solana dApps, including a Solidity DEX with liquidity pools, an on-chain game, and a Rust/Anchor social network utilizing TypeScript and IPFS.

Research on Blockchain Consensus Protocols (Master's Thesis)
- Implemented blockchain consensus protocols for Tezos, Proof of Work, and Proof of Authority. Created a modular consensus library and live testing tools using OCaml to evaluate performance metrics like TPS and block times.

On-disk Key Value Database
- Engineered a key-value disk database in C, utilizing B-trees for optimized data management and access.

Language Design / Compiler
- Coded a Go-based Lisp interpreter, and an OCaml-based Pascal to Assembly (X_86) compiler.

Skills

Programming	Rust, C/C++, Go, Python, Solidity, TypeScript/JavaScript
Technical Skills	Linux, Git, Web Development, Libp2p, CI/CD, Containerization, Databases (SQL/NoSQL), REST APIs, Data Structures & Algorithms
Languages	Portuguese (native speaker), English (fluent)

Interests

Technical Interests
Distributed Systems, Consensus Algorithms, Blockchains, Cryptography, Functional Programming, Compilers, Formal Verification, Operating Systems, Computer Security

Hobbies & Past Activities
Music (Saxophone, 5 yrs), Basketball (3 yrs), Fishing, Beekeeping, Farming, Cycling

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Awards and Events

Solana Breakpoint Conferences 2022 and 2023 **November 2023**
Received Student Fellowship Ticket @ Lisbon, Portugal/Amesterdam, Netherlands

ICPC SWERC 2023 **February 2023**
South Eastern Europe Programming Competition @ Politecnico Milano, Milan, Italy

MIUP 2022 **October 2022**
Portuguese National Programming Competition @ Coimbra University, Coimbra, Portugal